

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **GTM-13** **Date: 07.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A stationary particle breaks into two parts of masses m_A and m_B which move with velocity v_A and v_B respectively. The ratio of their kinetic energy ($K_B : K_A$) is

   A) $v_B : v_A$ B) $m_B : m_A$ C) $m_B v_B : m_A v_A$ D)1:1

2. Two planets A and B having masses m_1 and m_2 move around the sun in circular orbits of r_1 and r_2 radii respectively. If angular momentum of A is L and that of B is 3L, the ratio of time period $\left(\frac{T_A}{T_B}\right)$ is

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A) $\left(\frac{r_1}{r_2}\right)^3$ B) $\left(\frac{r_2}{r_1}\right)^{\frac{3}{2}}$ C) $\frac{1}{27}\left(\frac{m_2}{m_1}\right)^3$ D) $27\left(\frac{m_1}{m_2}\right)^3$

3. A mixture of one mole of monoatomic gas and one mole of a diatomic gas (rigid) are kept at room temperature (27°C). The ratio of specific heat of gases at constant volume respectively is

A) $\frac{3}{2}$ B) $\frac{5}{3}$ C) $\frac{7}{5}$ D) $\frac{3}{5}$

4. A player caught a cricket ball of mass 150g moving at a speed of 20*m/s*. If the catching process is completed in 0.1s, the magnitude of force exerted by the ball on the hand of the player is

A) 3 N B) 30 N C) 150 N D) 300 N

5. A series LCR circuit is at resonance for a capacitor C, inductance L and resistance R Now the value of resistance is halved keeping all other parameters same. The current amplitude at resonance will be now

A) Zero B) same C) double D) halved

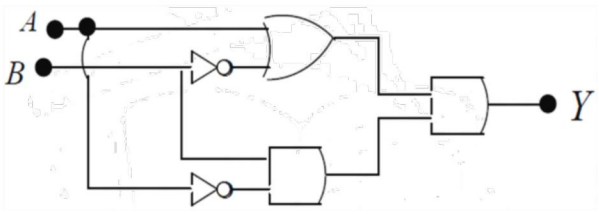
6. Average force exerted on a non-reflecting surface at normal incidence is 2.4×10^{-4}N. If 360W/cm^2 is the light energy flux during span of 1hour 30minutes. Then the area of the surface is

A) 20m^2 B) 0.02m^2 C) 0.1m^2 D) 0.2m^2

7. Binding energy of a certain nucleus is 18×10^8J. How much is the difference between total mass of all the nucleons and nuclear mass of the given nucleus

A) 2 μg B) 0.2 μg C) 20 μg D) 10 μg

8. The output Y of following circuit for given input is

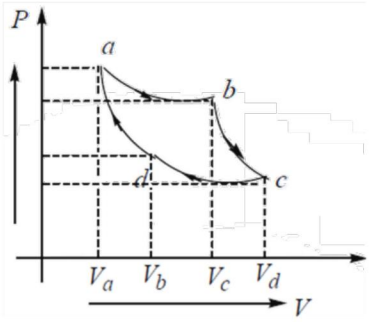


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A) $\bar{A}.B$ B) $A.B\ (A+B)$ C) 0 D) $A.B$

9. Two different adiabatic paths for the same gas interest two isothermal curves as shown in P-V diagram. The relation between the ratio $\frac{V_a}{V_d}$ and the ratio $\frac{V_b}{V_c}$



A) $\frac{V_a}{V_d}=\frac{V_b}{V_c}$ B) $\frac{V_a}{V_d}\neq\frac{V_b}{V_c}$ C) $\frac{V_a}{V_d}=\left(\frac{V_b}{V_c}\right)^2$ D) $\frac{V_a}{V_d}=\left(\frac{V_b}{V_c}\right)^{-1}$

10. Paramagnetic substance

a. align themselves along the directions of external magnetic field

b. attract strongly towards external magnetic field

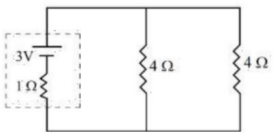
c. has susceptibility little more than zero

d. move from a region of strong magnetic field to weak magnetic field.

Choose the most appropriate answer from the options given below

A) a, c only B) a, b, c, d C) b, d only D) a, b, c only

11. In the given circuit the terminal potential difference of the cell is



Figure

A) 1.5 V B) 2V C) 3V D) 4V

12. Three bodies A, B and C have equal kinetic energies and their masses are 400g,1.2 kg and 1.6 kg respectively. The ratio of their linear momenta is

A) $\sqrt{3}:\sqrt{2}$ B) $\sqrt{3}:\sqrt{2}:1$ C) $\sqrt{2}:\sqrt{3}:1$ D) $1:\sqrt{3}:2$

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13. Critical angle of incidence for a pair of optical media is 45°. The refractive indices of first and second media are in the ratio

A) 2:1 B) $1:\sqrt{2}$ C) 1:2 D) $\sqrt{2}:1$

14. Young's modulus is determined by the equation given by $Y = 49000\frac{m}{l}\frac{dyne}{cm^2}$. where m is the mass and l is the extension of wire used in the experiment. Now error in Young modulus (Y) is estimated by taking data from $m - l$ plot in graph paper. The smallest scale divisions are 5 g and 0.02 cm along load axis and extension axis respectively. If the value of m and l are 500g and 2 cm respectively. then percentage error Y is

A) 2% B) 0.5% C) 0.02% D) 0.2%

15. In an expression $a \times 10^b$

A) b is order of magnitude for $a \leq 5$ B) b is order of magnitude for $5 < a \leq 10$

C) b is order of magnitude for $b \leq 5$ D) b is order of magnitude for $a \geq 5$

16. A proton and an electron are associated with same de-Broglie wavelength. The ratio of their kinetic energies is

(Assume h = 6.63×10^{-34} Js, $m_e = 9.0 \times 10^{-31} kg$ and m_p = 1836 times m_e)

A) $\frac{1}{1836}$ B) $\sqrt{1836}$ C) 1836 D) $\frac{1}{\sqrt{1836}}$

17. Correct Bernoulli's equation is (symbols have their usual meaning)

A) $P + \rho gh + \frac{1}{2}\rho v^2 = cons\tan t$ B) $P + \frac{1}{2}\rho gh + \frac{1}{2}\rho v^2 = cons\tan t$

C) $P + \rho gh + \rho v^2 = cons\tan t$ D) $P + mgh + \frac{1}{2}mv^2 = cons\tan t$

18. A clock has 75 cm, 60 cm long second hand and minute hand respectively. In 30 minutes duration the tip of second hand will travel x distance more than the tip of minute hand. The value of x in meter is nearly (Take $\pi = 3.14$)

A) 139.4 B) 220.0 C) 140.5 D) 118.96

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19. The diameter of a sphere is measured using a Vernier caliper whose 9 division of main scale are equal to 10 division of Vernier scale. The shortest division on the main scale is equal to 1mm. The main scale reading is 2 cm and second division of Vernier scale coincides with a division on main scale. If mass of the sphere is 8.635 g. the density of the sphere is

A) $2.2\ g/cm^3$ B) $1.7\ g/cm^3$ C) $2.5\ g/cm^3$ D) $2.0\ g/cm^3$

20. Two charged conducting sphere of radii a and b are connected to each other by a conducting wire. The ratio of charges of the two spheres respectively is

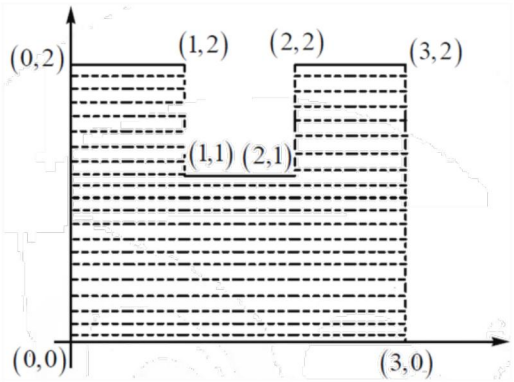
A) ab B) $\frac{b}{a}$ C) $\frac{a}{b}$ D) $\sqrt{ab}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. Resistance of a wire at 0°C,100°C and t°C is found to be 10Ω,10.2Ω and 10.95Ω respectively. The temperature t in Kelvin scale is___

22. A uniform thin metal plate of mass 10 kg with dimensions is shown. The ratio of x and y coordinates of center of mass of plate is $\frac{n}{9}$. The value of n is ___

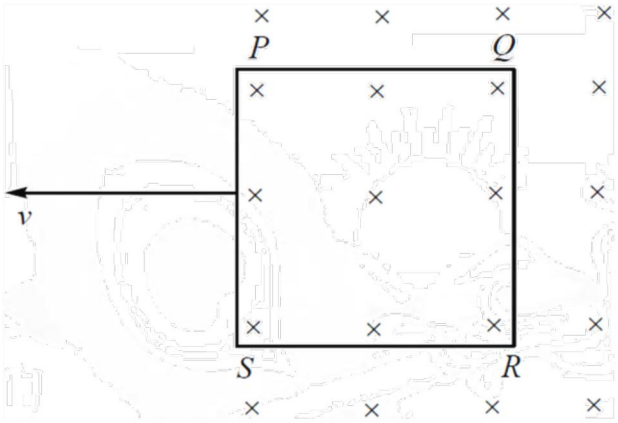


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23. A closed and an open organ pipe have same lengths. If the ratio of frequencies of their seventh overtones is $\left(\frac{a-1}{a}\right)$ then the value of a is ___

24. A square loop PQRS having 10 turns area 3.6×10^{-3} m^2 and resistance 100Ω is slowly and uniformly being pulled out of a uniform magnetic field of magnitude B = 0.5T as shown. Work done in pulling the loop out of the field in 1s is _____$\times10^{-6}$ J (Nearest integer)



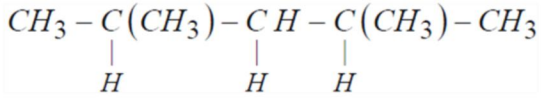
25. A liquid column of height 0.04 cm balance excess pressure of a soap bubble of certain radius. If density of liquid is 8×10^3 $kg\ m^{-3}$ and surface tension of soap solution is $0.28Nm^{-1}$, then diameter of the soap bubble is ____cm (if g = $10ms^{-2}$)

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. In the given compound, the number of 2° carbon atom/s is_____



A) Two B) One C) Three D) Four

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27. **Match List-I with List II**

Choose the correct answer from the options given below

	List I (Name of the test)		**LIST II (Reaction sequence involved) (M is metal)**
a.	Borax bead test	I	$MCO_3 \rightarrow MO \xrightarrow[+\Delta]{Co(NO_3)_2} CoOMO$
b.	Charcoal cavity test	II	$MCO_3 \rightarrow MCl_2 \rightarrow M^{2+}$
c.	Cobalt nitrate test	III	$MSO_4 \xrightarrow[\Delta]{Na_2B_4O_7} M(BO_2)_2 \rightarrow MBO_2 \rightarrow M$
d.	Flame test	IV	$MSO_4 \xrightarrow[\Delta]{Na_2CO_3} MCO_3 \rightarrow MO \rightarrow M$

A) $a-III, b-I, c-IV, d-II$ B) $a-III, b-I, c-II, d-IV$

C) $a-III, b-IV, c-I, d-II$ D) $a-III, b-II, c-IV, d-I$

28. Among the following halogens F_2, Cl_2, Br_2 and I_2 Which can undergo disproportionation reactions?

A) Cl_2, Br_2 and I_2 B) F_2 and Cl_2 C) F_2, Cl_2 and Br_2 D) Only I_2

29. Combustion of glucose ($C_6H_{12}O_6$) produces CO_2 and water. The amount of oxygen (in g) required for the complete combustion of 900 g of glucose is [Molar mass of glucose in $g\ mol^{-1}$ = 180]

A) 800 B) 480 C) 32 D) 960

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30. Match List-I with List II

	LIST I (Compound)		LIST II (Colour)
A	$Fe_4\left[Fe(CN)_6\right]_3 xH_2O$	I	Violet
B	$\left[Fe(CN)_5 NOS\right]^{4-}$	II	Blood Red
C	$\left[Fe(SCN)\right]^{2+}$	III	Prussian Blue
D	$(NH_4)_3 PO_4 12MoO_3$	IV	Yellow

Choose the correct answer from the options given below

A) A-IV, B-I, C-II, D-III

B) A-I, B-II, C-III, D-IV

C) A-III, B-I, C-II, D-IV

D) A-II, B-III, C-IV, D-I

31. Match List-I with List II

	LIST I (Elements)		LIST II (Properties in their respective groups)
a.	Cl, S	I	Elements with highest electronegativity
b.	Ge, As	II	Elements with larger atomic size
c.	Fr, Ra	III	Elements which show properties of both metals and non-metal
d.	F, O	IV	Elements with highest negative electron gain enthalpy

Choose the correct answer from the options given below

A) $a-III,\ b-II,\ c-I,\ d-IV$

B) $a-II,\ b-III,\ c-IV,\ d-I$

C) $a-II,\ b-I,\ c-IV,\ d-III$

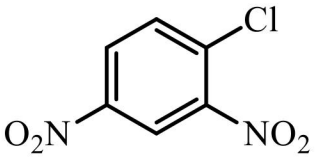
D) $a-IV,\ b-III,\ c-II,\ d-I$

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32. Given below are two statements

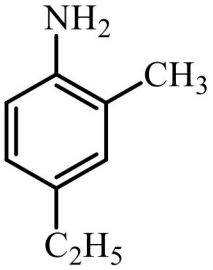
Statement-I:



Compound A

IUPAC name of compound A is 4-Chloro-1, 3-dinitrobenzene

Statement-II:



Compound B

IUPAC name of compound B is 4-Ethyl-2-methylaniline

In the light of the above statements choose the most appropriate answer from the options given below

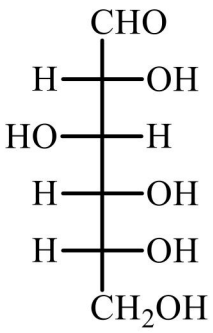
A) Both Statement I and Statement II are incorrect

B) Statement I is incorrect but Statement II is correct

C) Both Statement I and Statement II are correct

D) Statement I is correct but Statement II is incorrect

33. The incorrect statement regarding the given structure is



A) Will coexist in equilibrium with 2 other cyclic structure

B) can be oxidized to a dicarboxylic acid with Br_2 water

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C) despite the presence of – *CHO* does not gives Schiff's test

D) has 4 asymmetric carbon atoms

34. Iron (III) catalyses the reaction between iodide and persulphate ions, in which correct statements are:

a) Fe^{+3} oxidises the iodide ion

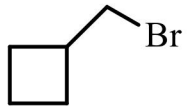
b) Fe^{+3} oxidises the persulphate ion

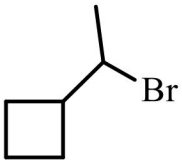
c) Fe^{2+} reduces the iodide ion

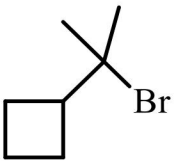
d) Fe^{2+} reduces the persulphate ion

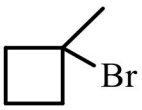
A) b & c only B) a only C) a & d only D) B only

35. Which among the following compound will undergo fastest SN^2 reaction

A) 

B) 

C) 

D) 

36. Given below are two statements

Statement I: $N(CH_3)_3$ and $P(CH_3)_3$ can act as ligands to form transition metal complexes

Statement II: As N and P are from same group, the nature of bonding of $N(CH_3)_3$ and $P(CH_3)_3$ is always same with transition metals

In the light of the above statements, choose the most appropriate answer from the options given below

A) Both Statement I and Statement II are incorrect

B) Statement I is incorrect but Statement II is correct

C) Both Statement I and Statement II are correct

D) Statement I is correct but Statement II is incorrect

37. Given below are two statements one is labelled as Assertion A and the other is labelled as Reason R

Assertion: The stability order of +1 oxidation state of Ga, *In* and *Tl* is $Ga < In < Tl$

Reason: The inert pair effect stabilizes the lower oxidation state down the group

In the light of the above statements, choose the correct answer from the options given

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below

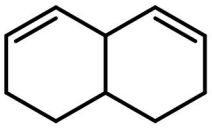
A) A is true but R is false

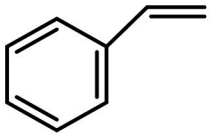
B) Both A and R are true and R is the correct explanation of A

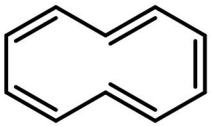
C) A is false but R is true

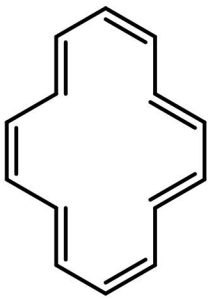
D) Both A and R are true but R is NOT the correct explanation of A

38. Which of the following are aromatics

a) 

b) 

c) 

d) 

A) Bond b only B) a and b only C) b and d only D) c and d only

39. For the given hypothetical reactions. The equilibrium constants are as follows

$X \rightleftharpoons Y; K_1 = 1.0$

$Y \rightleftharpoons; K_2 = 2.0$

$Z \rightleftharpoons W; K_3 = 4.0$

The equilibrium constant for the reaction $X \rightleftharpoons W$ is

A) 12.0 B) 8.0 C) 7.0 D) 6.0

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40. Match List-I with List II

	LIST I (Molecule)		**LIST II (Shape)**
a)	NH_3	I	Square pyramid
b)	BrF_5	II	Tetrahedal
c)	PCl_5	III	Traigonal pyramidal
d)	CH_4	IV	Trigonal bipyramidal

Choose the correct answer from the options given below

A) $a-IV, b-III, c-I, d-II$ B) $a-III, b-IV, c-I, d-II$

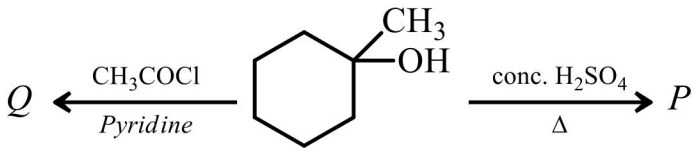
C) $a-III, b-I, c-IV, d-II$ D) $a-II, b-IV, c-I, d-III$

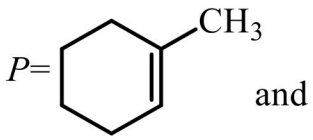
41. Number of Complexes with even number of electrons in t_{2g} orbitals is

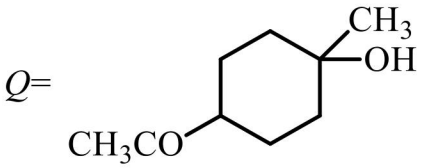
$\left[Fe(H_2O)_6\right]^{2+}, \left[Co(H_2O)_6\right]^{2+}, \left[Co(H_2O)_6\right]^{3+}, \left[Cu(H_2O)_6\right]^{2+}, \left[Cr(H_2O)_6\right]^{2+}$

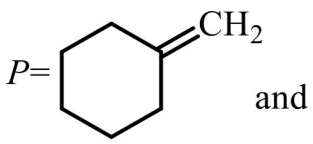
A) 3 B) 2 C) 5 D) 1

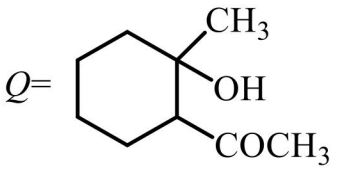
42. Identify the major product A and B respectively in the following set of reactions



A) 

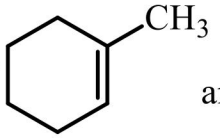
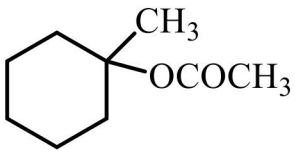
and 

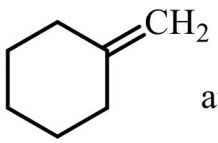
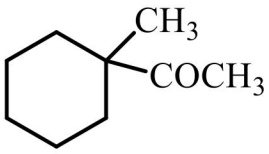
B) 

and 

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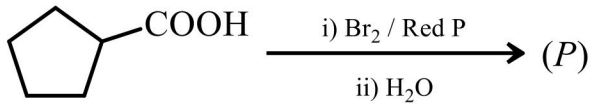
C) $P=$  and $Q=$ 

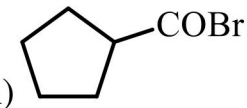
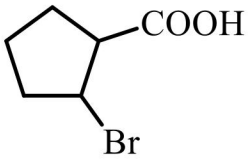
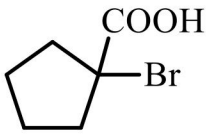
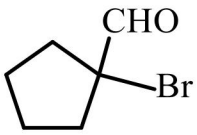
D) $P=$  and $Q=$ 

43. One mole of an octahedral complex with the formula $CoCl_3.nNH_3$ upon reaction with excess of $AgNO_3$ solution give 2 moles of $AgCl$ consider the oxidation state of Co in the complex is 'x'. The value of "x + n" is___

A) 8 B) 5 C) 6 D) 3

44. Identify the product (P) in the following reaction



A)  B)  C)  D) 

45. Thiosulphate reacts differently with iodine and bromine in the reactions given below

$2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$

$S_2O_3^{2-} + 4Br_2 + 5H_2O \rightarrow 2SO_4^{2-} + 8Br^- + 10H^+$

Which of the following statement justifies the above dual behaviour of thiosulphate?

A) Bromine undergoes oxidation and iodine undergoes reduction in these reactions

B) Thiosulphate undergoes oxidation by bromine and reduction by iodine in these reactions

C) Bromine is a stronger oxidant than iodine

D) Bromine is a weaker oxidant than iodine

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. If 279 g of aniline is reacted with one mole of benzenediazonium chloride. The maximum amount of yellow coloured dye formed will be _____g (nearest integer) (consider 100% yield of product)

47. A solution containing 10 g of an electrolyte AB_2 in 100g of water boils at 100.52°C. The degree of ionization of the electrolyte (α) is_____ $\times 10^{-1}$

[Given: Molar mass of AB_2 200 $gmol^{-1}$, k_b(molal boiling point elevation const. of water) = 0.52K $kg\ mol^{-1}$, boiling point of water = 100°C : $AB_{2(s)}$ ionises as $AB_{2(s)} \rightleftharpoons A^{2+}_{(aq)} + 2B^{-}_{(aq)}$]

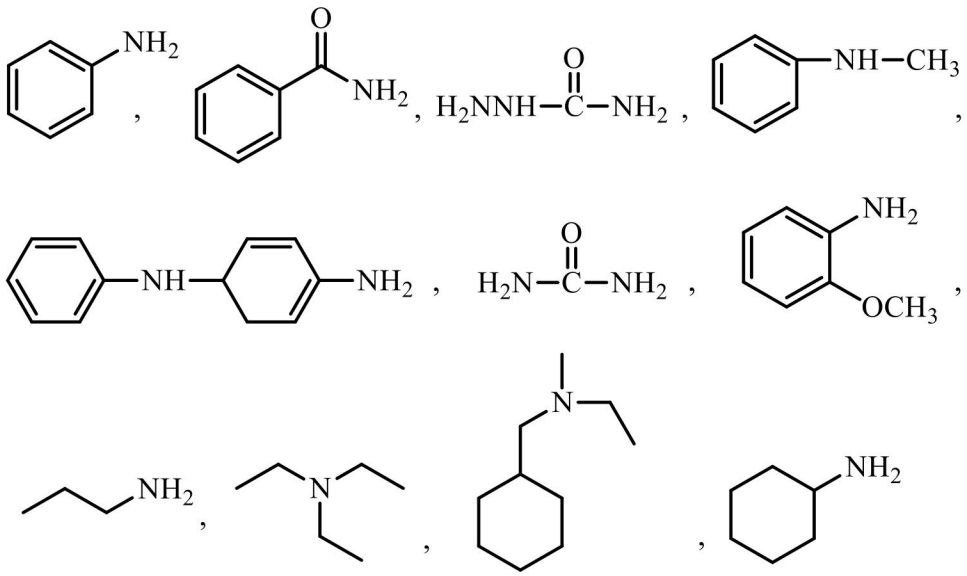
48. Consider the following reaction A + B → C

The time taken for A to become 1/4th of its initial concentration is twice the time taken to become 1/2 of the same. Also, when the concentration of B is plotted against time, the resulting graph gives a straight line with a negative slope and a positive intercept on the concentration axis. The overall order of the reaction is_____

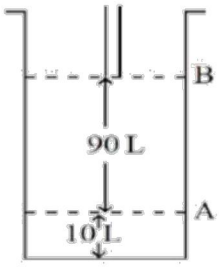
49. Number of amine compounds from the following giving solids which are soluble in NaOH upon reaction with Hinsberg's reagent is _____

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50. Consider the figure provided



1 mole of an ideal gas is kept in a cylinder fitted with a position, at the position A. at 18°C. If the position is moved to position B, keeping the temperature unchanged. Then 'x' L atm work is done in thus reversible process, $|x|=$___ L atm (nearest integer) (Given: Absolute temperature = °C + 273.15, $R = 0.08206 L\ atm\ mol^{-1}\ K^{-1}$)

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MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let the circle touch each $C_1:(x-\alpha)^2+(y-\beta)^2=r_1^2$ and $C_2:(x-8)^2+\left(y-\frac{15}{2}\right)^2=r_2^2$ touch each other externally at the point (6, 6). If the point (6, 6) divide the line segment joining the centers of the circles C_1 and C_2 internally in the ratio 2:1, then $(\alpha+\beta)+4\left(r_1^2+r_2^2\right)$ equals

A) 145 B) 130 C) 110 D) 125

52. Let $I(x)=\int\frac{6}{\sin^2 x(1-\cot x)^2}dx$. If $I(0)=3$, then $I\left(\frac{\pi}{12}\right)$ is equal to

A) $6\sqrt{3}$ B) $2\sqrt{3}$ C) $\sqrt{3}$ D) $3\sqrt{3}$

53. The number of critical points of the function $f(x)=(x-2)^{2/3}(2x+1)$ is

A) 2 B) 3 C) 1 D) 0

54. If the set $R=\{(a,b):a+5b=42,a,b\in N\}$ has m elements and $\sum_{n=1}^{m}\left(1-i^{n!}\right)=x+iy$, where $i=\sqrt{-1}$, then the value of $m+x+y$ is

A) 12 B) 5 C) 4 D) 8

55. If the shortest distance between lines

$L_1:\vec{r}=(2+\lambda)\hat{i}+(1-3\lambda)\hat{j}+(3+4\lambda)\hat{k}, \quad \lambda\in R$

$L_2:\vec{r}=(1+\mu)\hat{i}+3(1+\mu)\hat{j}+(5+\mu)\hat{k}, \quad \mu\in R$

Is $\frac{m}{\sqrt{n}}$, where $gcd(m, n) = 1$, then the value of $m + n$ equals

A) 387 B) 377 C) 384 D) 390

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56. Let [t] be the greatest integer less than or equals to t, Let A be the set of all prime factors of 2310 and $f:A\to Z$ be the function $f(x)=\left[\log_2\left(x^2+\left[\frac{x^3}{5}\right]\right)\right]$. The number of one to one functions from A to the range of f is

A) 24 B) 25 C) 120 D) 20

57. Let $f(x)$ be a positive function such that the area bounded by $y=f(x)$, $y=0$ from $x=0$ to $x-a>0$ is $e^{-a}+4a^2+a-1$. Then the differential equation, whose general solution is $y=c_1 f(x)+c_2$. where c_1 and c_2 are arbitrary constants is

A) $\left(8e^x-1\right)\frac{d^2y}{dx^2}-\frac{dy}{dx}=0$ B) $\left(8e^x+1\right)\frac{d^2y}{dx^2}-\frac{dy}{dx}=0$

C) $\left(8e^x+1\right)\frac{d^2y}{dx^2}+\frac{dy}{dx}=0$ D) $\left(8e^x-1\right)\frac{d^2y}{dx^2}+\frac{dy}{dx}=0$

58. Let $f(x)=4\cos^3 x+3\sqrt{3}\cos^2 x-10$. The number of points of local maxima of f in interval $(0,2\pi)$ is

A) 3 B) 4 C) 2 D) 1

59. Let z be a complex number such that $|z+2|=1$ and $lm\left(\frac{z+1}{z+2}\right)=\frac{1}{5}$. Then the value of $\left|\mathrm{Re}\left(\overline{z+2}\right)\right|$ is

A) $\frac{2\sqrt{6}}{5}$ B) $\frac{24}{5}$ C) $\frac{\sqrt{6}}{5}$ D) $\frac{1+\sqrt{6}}{5}$

60. Let $A=\begin{bmatrix}2 & a & 0\\ 1 & 3 & 1\\ 0 & 5 & b\end{bmatrix}$. If $A^3=4A^2-A-21I$. where I is the identity matrix of order 3×3, then $2a+3b$ is equal to

A) -9 B) -13 C) -10 D) -12

61. For the function $f(x)=\cos x-x+1, x\in R$, between the following two statements

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(S1): f (x) = 0 for only one value of x in $[0,\pi]$

(S2): f (x) is decreasing in $\left[0,\frac{\pi}{2}\right]$ and increasing in $\left[\frac{\pi}{2},\pi\right]$

A) only (S2) is correct
B) Both (S1) and (S2) are correct
C) only (S1) is correct
D) Both (S1) and (S2) are incorrect

62. Let the area of the region enclosed by the curve $y = \min(\sin x, \cos x)$ and the x-axis between $x = -\pi$ to $x = \pi$ be A, Then A^2 is equal to___

A) 36 B) 25 C) 16 D) 20

63. Let $H:\frac{-x^2}{a^2}+\frac{y^2}{b^2}=1$ be the hyperbola, whose eccentricity is $\sqrt{3}$ and the length of the latus return is $4\sqrt{3}$. Suppose the point $(\alpha,6),\alpha>0$ lies on H. If β is the product of the focal distances of the point $(\alpha,6)$. Then $\alpha^2+\beta$ is equal to

A) 171 B) 169 C) 172 D) 170

64. The set of all α, for which the vectors $\vec{a}=\alpha t\hat{i}+6\hat{j}-3\hat{k}$ and $\vec{b}=t\hat{i}-2\hat{j}-2\alpha t\hat{k}$ are inclined at an obtuse angle for all $t\in R$, is

A) $\left(-\frac{4}{3},0\right)$ B) $(-2,0]$ C) $[0,1)$ D) $\left(-\frac{4}{3},1\right)$

65. Let the sum of two positive integers be 24. If the probability that their product is not less than 3/4 times their greatest possible product is $\frac{m}{n}$, where $gcd\,(m,n) = 1$, then $n - m$ is equal to

A) 9 B) 11 C) 8 D) 10

66. If $\sin x=-\frac{3}{5}$, where $\pi<x<\frac{3\pi}{2}$. then $80\left(\tan^2 x-\cos x\right)$ is equal to

A) 19 B) 108 C) 18 D) 109

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67. Ley $p(x, y, z)$ be a point in the first octant. Whose projection in the xy – plane is the point Q. Let $OP = \gamma$: the angle between OQ and the positive x-axis be θ and the angle between the OP and the positive z-axis be ϕ. Where O is the origin, Then the distance of P from the x-axis is

A) $\gamma\sqrt{1+\cos^2\theta\sin^2\phi}$ B) $\gamma\sqrt{1-\sin^2\theta\cos^2\phi}$

C) $\gamma\sqrt{1-\sin^2\phi\cos^2\theta}$ D) $\gamma\sqrt{1+\cos^2\phi\sin^2\theta}$

68. The value of $k \in N$ for which the integral $I_n = \int_0^1 \left(1-x^k\right)^n dx, n \in N$ satisfies $147I_{20} = 148I_{21}$ is

A) 7 B) 8 C) 10 D) 14

69. Let $y = y(x)$ be the solution of the differential equation $\left(1+y^2\right)e^{\tan x}dx+\cos^2 x\left(1+e^{2\tan x}\right)dy=0,$

$y(0)=1$. Then $y\left(\frac{\pi}{4}\right)$ is equal to

A) $\frac{2}{e}$ B) $\frac{1}{e}$ C) $\frac{1}{e^2}$ D) $\frac{2}{e^2}$

70. The equations of two sides AB and AC of a triangle ABC are $4x + y = 14$ and $3x – 2y = 5$, respectively. The point $\left(2,-\frac{4}{3}\right)$ divides the third side BC internally in the ratio 2:1. the equation of the side BC is

A) $x – 6y – 10 = 0$ B) $x + 6y + 6 = 0$ C) $x + 3y + 2 = 0$ D) $x – 3y – 6 = 0$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. Three balls are drawn at random from a bag containing 5 blue and 4 yellow balls. Let the random variable X and Y respectively denote the number of blue and yellow balls. If $\overline{X}$ and $\overline{Y}$ are the mean of X and Y respectively. Then $7\overline{X}+4\overline{Y}$ is equal to__

72. Let the positive integer be written in the form

1

2 **3**

4 **5** **6**

7 **8** **9** **10**

If the kth row contains exactly k numbers for every natural number k, then the row in which the numbers 5310 will be is___

73. Let $\alpha=\sum_{r=0}^{n}\left(4r^2+2r+1\right){}^nC_r$ and $\beta=\left(\sum_{r=0}^{n}\frac{{}^nC_r}{r+1}\right)+\frac{1}{n+1}$. If $140<\frac{2\alpha}{\beta}<281$ then the value of n is

74. The value of $\lim_{x\to0}2\left(\frac{1-\cos x\sqrt{\cos 2x}\sqrt[3]{\cos 3x}....\sqrt[10]{\cos 10x}}{x^2}\right)$ is _____

75. If the range of $f(\theta)=\frac{\sin^4\theta+3\cos^2\theta}{\sin^4\theta+\cos^2\theta}$, $\theta\in R$ is $[\alpha,\beta]$, then the sum of the infinite G.P. whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$ is equal to

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